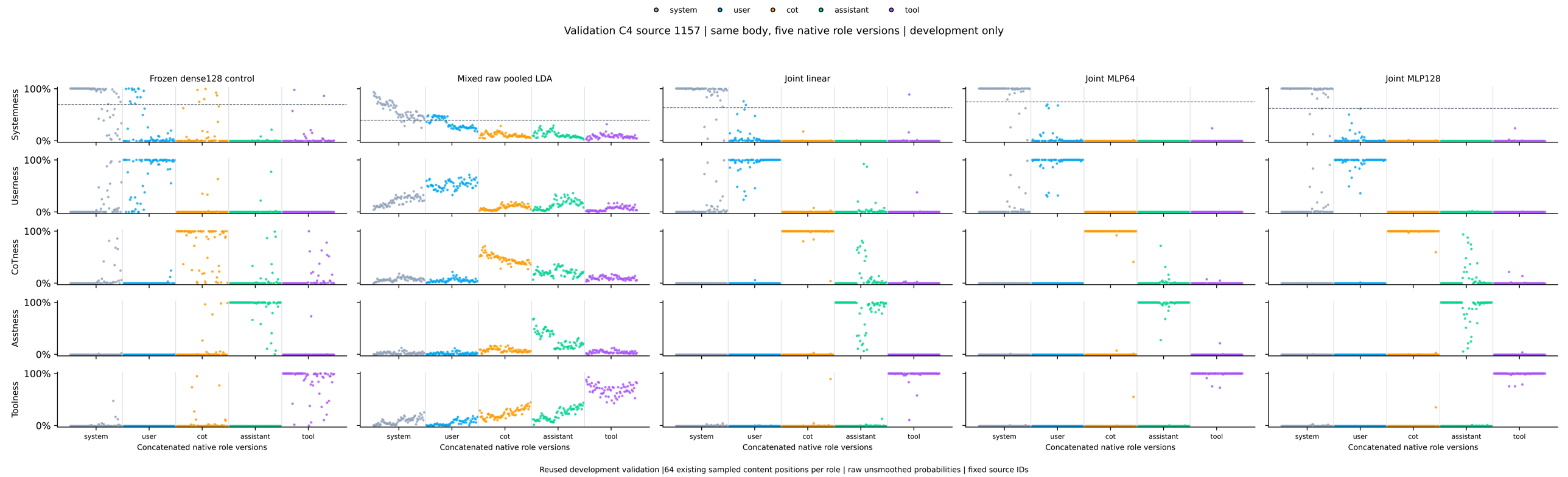

system
user
cot
assistant
tool
Validation C4 source 1157 | same body, five native role versions | development only
Frozen dense128 control
Mixed raw pooled LDA
Joint linear
Joint MLP64
Joint MLP128
Systemness
Userness
CoTness
Asstness
Toolness
100%
0%
Concatenated native role versions
Reused development validation |64 existing sampled content positions per role | raw unsmoothed probabilities | fixed source IDs